

Layton Sahler

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PROFESSIONAL SUMMARY

Computer and Electrical Engineering student at Boston University with hands-on experience across embedded firmware, hardware design, and full-stack web development. Built award-winning medical wearables, real-time signal processing systems, and production research tools used in classroom instruction. Comfortable working across the full stack from KiCad schematics to Neo4j graph databases and React front ends.

EDUCATION

Boston University

BS, Computer Engineering and Electrical Engineering. Dean's List. GPA: 3.70.

May 2028

Relevant Coursework: Cybersecurity, Computer Networking, Object Oriented Programming, Algorithms, Logic Design, Electrical Circuits, Linear Algebra, Multivariate and Differential Calculus

SKILLS

Programming Languages: C++, C, Python (NumPy, pandas, Matplotlib), JavaScript, MATLAB, SQL, Cypher

Web: HTML/CSS, React, Next.js, QT/QML

Tools & Platforms: Neo4j, MongoDB, Git, KiCad, Unity, Fusion 360, OnShape, Unix

PROJECTS

Dean's Imagiengineering Competition: VenoSense

- Designed VenoSense as sole engineer on a 2-person founding team: a wearable device screening postoperative patients for deep vein thrombosis, winning Best in Class (Sophomores)
- Built a cross-platform BLE application in C++/Qt/QML to visualize real-time ECG, flex sensor, and 3-axis accelerometer data streamed from hardware
- Engineered full embedded firmware on Arduino/ESP32, including I2C multi-sensor integration, signal conditioning, and BLE framing
- Designed power regulation and sensor conditioning circuitry in KiCad, producing a readable signal and enabling automatic heart rate detection via threshold-based peak analysis

Lutron Lighting Competition

C++, Microcontrollers, Laser Cutting

- Designed, soldered, and assembled a 70-pixel LED matrix to display real-time frequency band data
- Implemented Fast Fourier Transform (FFT) algorithms in C++ on a microcontroller for real-time audio analysis and visualization
- Modeled laser-cut acrylic enclosures and 3D-printed supports in OnShape, iterating through rapid prototyping cycles
- Led system integration, debugging, and testing as primary contributor, placing first against 21 competing teams

EXPERIENCE

Research Assistant

May 2026 – Present

BU

- Build a map/poem visualization page for a faculty research site on The Tale of Genji from a high-level spec, implementing a custom force-directed physics simulation for node placement with full pan/zoom/drag interactivity
- Authored a graph data model with 300+ node entities and 5,000+ relationships in Neo4j via Cypher
- Write multi-hop Cypher queries and REST API endpoints following the site's existing API conventions
- Implemented human-in-the-loop content verification for AI-generated vs. human-reviewed poem translations, adapted from the site's admin-review pattern and now used by researchers validating the database
- Iterated on design and functionality based on faculty feedback; page is now used in classroom instruction and as a research aid

Robotics Instructor

May 2025 – Aug 2025

ID TECH

- Managed sole responsibility for a classroom of students ages 12-17 across six week-long VEX BattleBots courses, overseeing curriculum pacing, safety, and student progress
- Mentored advanced students in algorithmic thinking, control logic, and robot software development beyond standard course material

STEM Instructor

Feb 2025 – May 2025

Brains and Motion

- Facilitated hands-on STEM lessons for a class of 30 students (ages 6-9), collaborating with a team of four instructors to design activities and support individual learning